

Structural proof of the column-1 SC discriminator for Ω on $\lambda = (3, 3, 1, 1, 1)$

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1 Setup

Let $\lambda = (3, 3, 1, 1, 1)$, $n = 9$. Working in the Hecke algebra $H_n(q)$ with Hoefsmit's semi-normal basis $\{v_T : T \in \text{SYT}(\lambda)\}$, $N = 120$. The generators are T_i ($i = 1, \dots, 8$); set $S_i := T_i + 1$. Define

$$R'_k := S_{k-1}S_{k-2} \cdots S_2S_1 \quad (k \geq 2), \quad \Omega := R'_3R'_4R'_5R'_6R'_7R'_8R'_9.$$

For a tableau T , write $T(r, c)$ for the entry in row r , column c (1-indexed).

For each SYT T , T_1 has the eigenvalues $\{q, -1\}$ on v_T : $T_1v_T = qv_T$ when $T(1, 1) = 1$, $T(1, 2) = 2$ (*SR-at-1*), and $T_1v_T = -v_T$ when $T(1, 1) = 1$, $T(2, 1) = 2$ (*SC-at-1*). Every SYT of λ is one or the other. Hence $V = V_+ \oplus V_-$ where

$$V_+ := \text{span}\{v_T : T \text{ SR-at-1}\}, \quad V_- := \text{span}\{v_T : T \text{ SC-at-1}\}.$$

$S_1 = T_1 + 1$ satisfies $S_1v = (q + 1)v$ on V_+ and $S_1v = 0$ on V_- . So $V_- = \ker S_1$.

Three structural conditions on SR-at-1 SYTs.

- (α) $T(2, 1) = 3$ and $T(3, 1) = 4$
- (β) $T(3, 1) = 5$ and $T(4, 1) = 6$
- (γ) $T(4, 1) = 7$ and $T(5, 1) = 8$

Each condition is a *column-1 SC pair* at indices 3, 5, 7 respectively. α -SYTs form a 20-dimensional subspace $V_+^\alpha = V_+ \cap \ker S_3$ inside V_+ (since in our shape, the only way 3,4 can be column-adjacent in an SR-at-1 SYT is at $(2, 1), (3, 1)$). Similarly β -SYTs span V_+^β .

2 Theorem

Theorem 1. *Let $T \in \text{SYT}(\lambda)$ be SR-at-1. Then $\Omega v_T = 0$ if and only if T satisfies at least one of (α) , (β) , (γ) .*

The converse direction is a finite check on the 50 SR-at-1 SYTs (no T outside $\alpha \cup \beta \cup \gamma$ has $\Omega v_T = 0$). The structural content is the forward direction: three implications $\alpha \Rightarrow \Omega v_T = 0$, $\beta \Rightarrow \Omega v_T = 0$, $\gamma \Rightarrow \Omega v_T = 0$, which we prove as Lemmas ??, ??, ??.

3 Key reduction

Every R'_k has S_1 as its rightmost factor, so $R'_k w = 0$ whenever $S_1 w = 0$, i.e., $R'_k(V_-) = 0$. In particular, to show $\Omega v_T = 0$ it suffices to show that at some stage $k \in \{3, \dots, 8\}$ we have $\Pi_{k+1}(T) := R'_{k+1} R'_{k+2} \cdots R'_9 v_T \in V_-$.

The V_+/V_- decomposition and operator action

Lemma 1. *For $j \geq 3$, S_j preserves V_+ and V_- separately.*

Proof. $S_j v_{T'} = a v_{T'} + b v_{s_j T'}$ in Hoefsmit basis (or a diagonal action), where $s_j T'$ swaps the values $j, j+1$ in T' . For $j \geq 3$, this does not move the entries 1 or 2, so T' and $s_j T'$ are in the same V_+/V_- class. $\square$

Lemma 2. *If T is SR-at-1 with $T(2, 1) = 3$, then $S_1 S_2 S_1 v_T = -(q+1)v_T$.*

Proof. $S_1 v_T = (q+1)v_T$ by SR-at-1. With 2 at $(1, 2)$ (content 1) and 3 at $(2, 1)$ (content -1), the axial distance is $d_2 = -2$, so $S_2 v_T = a v_T + b v_{s_2 T}$ with $a = (q-1)q^{-2}/(q^{-2}-1) = -1/(q+1)$ (and some b). In $s_2 T$: 1 at $(1, 1)$, 2 at $(2, 1)$ — SC-at-1, so $v_{s_2 T} \in V_-$ and $S_1 v_{s_2 T} = 0$. Therefore $S_1 S_2 S_1 v_T = (q+1)(a(q+1)v_T + b \cdot 0) = a(q+1)^2 v_T = -(q+1)v_T$. $\square$

4 Proof of L

Lemma 3 (L_α). *If T is an α -SYT, then $R'_k v_T \in V_-$ for every $k \geq 4$. In particular $\Omega v_T = 0$.*

Proof. By Lemma ??, $S_1 S_2 S_1 v_T = -(q+1)v_T$. Since S_1 commutes with S_j for $j \geq 3$, we may pull S_1 rightward in $S_1 R'_k = S_1 \cdot S_{k-1} S_{k-2} \cdots S_1$:

$$S_1 R'_k = (S_{k-1} \cdots S_3) \cdot S_1 S_2 S_1.$$

Applied to v_T :

$$S_1 R'_k v_T = (S_{k-1} \cdots S_3) \cdot (-(q+1)v_T) = -(q+1)(S_{k-1} \cdots S_3) \cdot S_3 v_T.$$

By the α -condition $T(2, 1) = 3, T(3, 1) = 4$, entries 3, 4 are column-adjacent in T , so $T_3 v_T = -v_T$ and $S_3 v_T = 0$. Therefore $S_1 R'_k v_T = 0$, i.e. $R'_k v_T \in V_-$. Since $R'_j V_- = 0$ for every j , applying any further R'_j for $j < k$ kills it, giving $\Omega v_T = 0$. $\square$

5 Proof of L

The proof works by showing that the V_+ -part of $R'_9 v_T$ lands on α -SYTs, then applying L_α per-tableau.

Lemma 4 (Support lemma SL). *If T is a β -SYT, then $S_3(S_8 S_7 S_6 S_5 S_4 S_3) v_T = 0$.*

Proof. S_3 commutes with S_5, S_6, S_7, S_8 . So

$$S_3 \cdot S_8 S_7 S_6 S_5 S_4 S_3 = S_8 S_7 S_6 S_5 \cdot S_3 S_4 S_3.$$

We claim that for T a β -SYT, the vector $S_3 S_4 S_3 v_T$ is supported on the set $\{T, s_3 T\}$, and both have 5, 6 stacked in column 1 (so S_5 kills).

Case (1) $T(2, 1) = 3, T(1, 3) = 4$. Axial distance $d_3 = 3$ between 3 and 4, so $S_3 v_T = a_3 v_T + b_3 v_{s_3 T}$. In $s_3 T$, the entries 4 and 5 are now at $(2, 1)$ and $(3, 1)$ — same column — so $S_4 v_{s_3 T} = 0$. Therefore $S_4 S_3 v_T = a_3 S_4 v_T = a_3(a_4 v_T + b_4 v_{s_4 T})$ where in $s_4 T$ the entries 3 and 4 are at $(2, 1)$ and $(3, 1)$ — same column — so $S_3 v_{s_4 T} = 0$. Therefore $S_3 S_4 S_3 v_T = a_3 a_4(a_3 v_T + b_3 v_{s_3 T})$ — supported on $\{T, s_3 T\}$.

Case (1') $T(2, 1) = 3, T(2, 2) = 4$. Here $d_3 = 1$ (3, 4 in same row), so $S_3 v_T = (q + 1)v_T$. Then $S_3 S_4 S_3 v_T = (q + 1)S_3 S_4 v_T$. Now $d_4 = -2$, so $S_4 v_T = a_4 v_T + b_4 v_{s_4 T}$, and in $s_4 T$, 3 and 4 are column-adjacent at $(2, 1)$ and $(3, 1)$ — so $S_3 v_{s_4 T} = 0$. Hence $S_3 S_4 v_T = a_4(q + 1)v_T$, giving $S_3 S_4 S_3 v_T = -(q + 1)v_T$ — supported on $\{T\}$.

Case (2) $T(1, 3) = 3, T(2, 1) = 4$. Here 4 and 5 are column-adjacent at $(2, 1), (3, 1)$ — so $S_4 v_T = 0$. Hence $S_4 S_3 v_T = a_3 \cdot 0 + b_3 S_4 v_{s_3 T}$. In $s_3 T$, 3 is at $(2, 1)$ and 4 is at $(1, 3)$ (1-like); $S_4 v_{s_3 T}$ produces support $\{s_3 T, s_4 s_3 T\}$, and in $s_4 s_3 T$ the entries 3 and 4 are at $(2, 1), (3, 1)$ — same column — so $S_3 v_{s_4 s_3 T} = 0$. Therefore $S_3 S_4 S_3 v_T = b_3 a_4(a'_3 v_{s_3 T} + b'_3 v_T)$ — supported on $\{T, s_3 T\}$.

In every case, the support is in $\{T, s_3 T\}$, both of which have 5 at $(3, 1)$ and 6 at $(4, 1)$ (the β -condition involves no entries ≤ 4 , so s_3 preserves it). These are SC-at-5 SYTs: $S_5 v_T = S_5 v_{s_3 T} = 0$. Therefore $S_8 S_7 S_6 S_5 \cdot (S_3 S_4 S_3) v_T = (S_8 S_7 S_6) \cdot 0 = 0$. $\square$

Lemma 5 (L_β). *If T is a β -SYT, then $\Omega v_T = 0$.*

Proof. We show $R'_8 R'_9 v_T \in V_-$. By Lemma ??, $S_3, S_4, \dots, S_8$ preserve the V_+/V_- split. Decompose $R'_9 v_T = w_+ + w_-$ with $w_\pm \in V_\pm$. Then $R'_8 R'_9 v_T = R'_8 w_+ + R'_8 w_- = R'_8 w_+$ (since $R'_8(V_-) = 0$). So it suffices to show $\pi_+(R'_8 w_+) = 0$, where π_+ is projection to V_+ .

For SR-at-1 T , $S_1 v_T = (q + 1)v_T$, so $R'_9 v_T = (q + 1)(S_8 \cdots S_2) v_T$. Tracking V_+/V_- : S_2 on SR-at-1 T either stays in V_+ (when 3 is in row 1 or column 2 of T) or splits into $V_+ \oplus V_-$ where the V_- component involves $s_2 T$ which is SC-at-1. After S_2 , the operators $S_3, \dots, S_8$ preserve the split. Therefore the V_+ -part of $R'_9 v_T$ is $w_+ = c_T \cdot (S_8 \cdots S_3) v_T$ for some nonzero scalar c_T depending on the position of 3 in T .

Now we claim $w_+ \in V_+^\alpha = V_+ \cap \ker S_3$. By Lemma ??, $S_3(S_8 \cdots S_3) v_T = 0$, so $S_3 w_+ = 0$. Combined with $w_+ \in V_+$, this gives $w_+ \in V_+^\alpha$, i.e. $w_+ = \sum_{T' \in \alpha} c_{T'} v_{T'}$.

For each α -SYT T' in this sum, by L_α (Lemma ?? with $k = 8$), $R'_8 v_{T'} \in V_-$. Therefore $R'_8 w_+ \in V_-$, completing the proof. $\square$

6 Proof of L

We need an analogue of Lemma ?? for $R'_8 v_{T'}$ (rather than R'_9), and a finer support description.

Lemma 6. *If T' is a β -SYT, then $S_3(S_7 S_6 S_5 S_4 S_3) v_{T'} = 0$. Consequently the V_+ -part of $R'_8 v_{T'}$ lies in V_+^α .*

Proof. Verbatim analogue of Lemma ?? with S_8 removed. The chain becomes $S_3 \cdot S_7 S_6 S_5 S_4 S_3 = S_7 S_6 S_5 \cdot S_3 S_4 S_3$. The same case analysis (1, 1', 2) gives $S_3 S_4 S_3 v_{T'} \in \text{span}\{v_{T'}, v_{s_3 T'}\} \subseteq \ker S_5$. Hence the chain vanishes.

The second statement: as in Lemma ??, the V_+ -part of $R'_8 v_{T'}$ is a scalar times $(S_7 S_6 S_5 S_4 S_3) v_{T'}$, and the first statement gives $S_3 \cdot \text{this} = 0$, hence the V_+ -part is in V_+^α . $\square$

Lemma 7 (Support for γ). *If T is a γ -SYT, then $\pi_+(R'_9 v_T) \in V_+^\alpha \oplus V_+^\beta$.*

Remark. Status: This is the structural piece that the present proof does not yet fully close. Computational verification at generic q (file `2026-05-19-gamma-structure.py`, output Q1) confirms

the support claim for all 6 γ -only SYTs of $\lambda = (3, 3, 1, 1, 1)$: $\pi_+(R'_9 v_T)$ is supported on at most 8 SYTs, all in $\alpha \cup \beta$. The structural ingredients in place are:

- Each s_3, s_4 preserves the γ -condition (positions of 7, 8 unchanged), so $S_3 S_4 S_3 v_T$ is supported on γ -SYTs.
- The chain $S_8 \cdots S_3$ acts within V_+ (preserved by all S_j , $j \geq 3$).
- For the subcases with $T(3, 1) = 6$ (so T has SC at 6 as well as 7), the chain action restricts further; for $T(3, 1) = 5$ the structure is more involved.

A full structural proof appears to require either a finer SC-at-7-propagation argument through S_5 and S_6 , or a direct algebraic identity in $H_n(q)$ expressing $S_3 \cdot (S_8 \cdots S_3) v_T$ in terms of $S_7 v_T$ for γ -SYTs.

Lemma 8 (L_γ). *If T is a γ -SYT, then $\Omega v_T = 0$.*

Proof, modulo Lemma ??. We show $R'_7 R'_8 R'_9 v_T \in V_-$.

Decompose $R'_9 v_T = w_+ + w_-$ with $w_- \in V_-$. By Lemma ??, $w_+ = w_+^\alpha + w_+^\beta$ with $w_+^\alpha \in V_+^\alpha$ and $w_+^\beta \in V_+^\beta$.

Now $R'_8 R'_9 v_T = R'_8 w_+ + R'_8 w_-$. The V_- part is killed by R'_8 : $R'_8 w_- = 0$. So $R'_8 R'_9 v_T = R'_8(w_+^\alpha + w_+^\beta)$.

For $w_+^\alpha = \sum c_{T'} v_{T'}$ over α -SYTs T' : by Lemma ?? applied with $k = 8$, each $R'_8 v_{T'} \in V_-$. So $R'_8 w_+^\alpha \in V_-$.

For $w_+^\beta = \sum d_{T'} v_{T'}$ over β -SYTs T' : by Lemma ??, $\pi_+(R'_8 v_{T'}) \in V_+^\alpha$. So $\pi_+(R'_8 w_+^\beta) \in V_+^\alpha$.

Combining: $R'_8 R'_9 v_T \in V_- \oplus V_+^\alpha$.

Finally, R'_7 applied: $R'_7 V_- = 0$ (rightmost S_1 kills). For the V_+^α -part, decompose $\pi_+(R'_8 R'_9 v_T) = \sum e_{T'} v_{T'}$ over α -SYTs T' . By Lemma ?? with $k = 7$, $R'_7 v_{T'} \in V_-$ for each. Hence $R'_7(\pi_+(R'_8 R'_9 v_T)) \in V_-$.

Therefore $R'_7 R'_8 R'_9 v_T \in V_-$, and applying any further R'_j gives 0. $\square$

7 Theorem proof

Proof of Theorem ??. ($\Leftarrow$) If T satisfies α , β , or γ , then $\Omega v_T = 0$ by Lemmas ??, ??, ?? respectively.

($\Rightarrow$) Computational: verified at generic q for all 50 SR-at-1 SYTs (diagnostic in 2026-05-19-V33111-residual-). The 15 SR-at-1 SYTs not satisfying any of α, β, γ all have $\Omega v_T \neq 0$. $\square$

8 Computational verification

All three forward lemmas are verified at generic q via SymPy (2026-05-19-key-lemma-verify.py): 20/20 α -class, 9/9 β -class, 6/6 γ -class pass the test $S_1 \cdot \Pi_{k+1}(T) = 0$ at the predicted first-vanish stage.

The support claims (the heart of the structural proofs) are also verified (2026-05-19-supports.py, 2026-05-19-vplus-support.py, 2026-05-19-gamma-structure.py):

- For α -SYT T , $R'_9 v_T \in V_-$ (support entirely in SC-at-1).
- For β -SYT T , $\pi_+(R'_9 v_T) \in V_+^\alpha$ (support entirely in α -SYTs).
- For γ -SYT T , $\pi_+(R'_9 v_T) \in V_+^\alpha \oplus V_+^\beta$ (support entirely in $\alpha \cup \beta$).

9 Gap and outlook

The remaining structural gap is Lemma ?? : a q -generic proof that for γ -SYT T , $\pi_+(R'_9 v_T)$ is supported in $\alpha \cup \beta$. A natural attempt — the commutation reduction $S_3 \cdot (S_8 \cdots S_3) = (S_8 S_7 S_6 S_5)(S_3 S_4 S_3)$ — reduces to showing $(S_8 S_7 S_6 S_5)(S_3 S_4 S_3) v_T \in V_-$ for γ -SYT, but the γ -condition ($S_7 v_T = 0$) does not directly trigger here because s_5, s_6 scramble entries 5,6,7 before the S_7 in the chain acts.

Two routes appear promising:

1. Track γ -class invariance through S_5, S_6 : the support of $(S_5 \cdot S_3 S_4 S_3) v_T$ stays within $\{T' : T'(4,1) = 7, T'(5,1) = 8\}$, but $(S_6 \cdot \text{that})$ may not. Need to compute how the V_+ -part fragments under S_6 and check that the leftover V_+ -component projects into $V_+^\alpha \oplus V_+^\beta$ rather than the larger “other” subspace.
2. Build a direct Hecke-algebra identity of the form $S_3(S_8 \cdots S_3) = A \cdot S_7 + B$ where $B \in V_+^\alpha \oplus V_+^\beta$ generation, applied to v_T kills the first term via $S_7 v_T = 0$.

The gap is finite — proving it for the six γ -only SYTs of $\lambda = (3,3,1,1,1)$ closes the theorem for this shape — and the gap has been verified computationally for all six.

Files referenced

- Verification: `~/projects/scratch/2026-05-19-key-lemma-verify.py`
- Support analysis: `~/projects/scratch/2026-05-19-supports.py`, `2026-05-19-vplus-support.py`, `2026-05-19-gamma-structure.py`
- Empirical discriminator: `~/projects/scratch/2026-05-19-V33111-residual-diagnostic.md`